Treatment of RA : A Clinical Pharmacy Perspective

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Pharmacological TTT:

Aims:

improve the move
break the ache
manage the damage

- A- First line drugs
  - NSAID
  - Corticosteroids
- B- second line drugs; DMARDs
  - Conventional
  - Biologics(targeted); biologic response modifiers; BRMs

NSAIDs

- 1st choice
- ↓ pain, inflammation, swelling
- Not affect disease progression
- Not enough alone
- SE: GIT upset

Corticosteroids:

- Immediate and dramatic relief
- Both immunosuppressive and anti-inflammatory
- Routes
  - Oral small dose (prednisone)
    - Bridging ttt until DMARDs act
  - IV
    - Severe cases (vasculitis)
  - IM
    - Acute flare
  - Intraarticular (triamcinolone)
    - 1 or 2 joints
    - Not > 4 times/yr to avoid painless destruction of joint

DMARDs

- Alter disease progression
- significantly changed prognosis of disease
- Most of them are immunosuppressive

Conventional DMARDs (cDMARDs)

broadly modulate immune activity (e.g., methotrexate inhibits folate metabolism)

- MTX
- Sulfasalazine
- Azathioprine
- Hydroxychloroquine
- Leflunomide
- Mycophenolate mofetil

- **Methotrexate (MTX):**
  - A folate analogue with anti-inflammatory properties
  - 1st line DMARD
  - Dose 15-25 mg/wk
  - Onset 2-4 wks
  - SE
    - Gastitis, stomatis → MC
    - BM suppression
    - Hepatotoxicity
- **Sulfasalazine:**
  - Suppresses B&T cells
  - ↓ release of cytokines
  - High continuation rates
  - SE:
    - Nasea
    - Headache, malaise
    - Neutropenia, ↓ PLT

- **Leflunomide**
  - inhibitor of pyrimidine synthesis → ↓ DNA & RNA → ↓ T cell activity
  - CI in preg
  - SE:
    - Rash
    - Alopecia
    - HTN
    - Hepatotox
    - teratogenic
- **Hydroxychloroquine**
  - Antimalarial
  - Blocks the activation of toll-like receptors (TLR), which decreases the activity of dendritic cells, thus mitigating the inflammatory process
  - SE
    - Retinitis
    - myopathy

Biologics/Targeted DMARDs.

block specific inflammatory signals (e.g., TNF, IL-1, IL-6).

- **T-cell suppression:** *Abatacept*.
- **B-cell depletion:** *Rituximab*.
- **Interleukin-1 (IL-1) blockade:** *Anakinra*.
- **Interleukin-6 (IL-6) blockade:** *Tocilizumab*.
- **Tumor Necrosis Factor (TNF) inhibitors:** *Adalimumab, etanercept, infliximab*.

- SE:
  - ↑ TB infection risk
  - Injection site irritation
  - infusion reactions
  - Interstitial pneumonia

Treatment Algorithm

- **Early RA:** MTX + short-term steroids.
- **Moderate-Severe RA:** MTX → add bDMARD/tsDMARD if inadequate response.
- **Personalized therapy:** Comorbidities, cost, patient preference.

Clinical Pharmacist’s Role

1. Medication reconciliation.
2. Adherence counseling.
3. Toxicity monitoring.
4. Multidisciplinary collaboration.